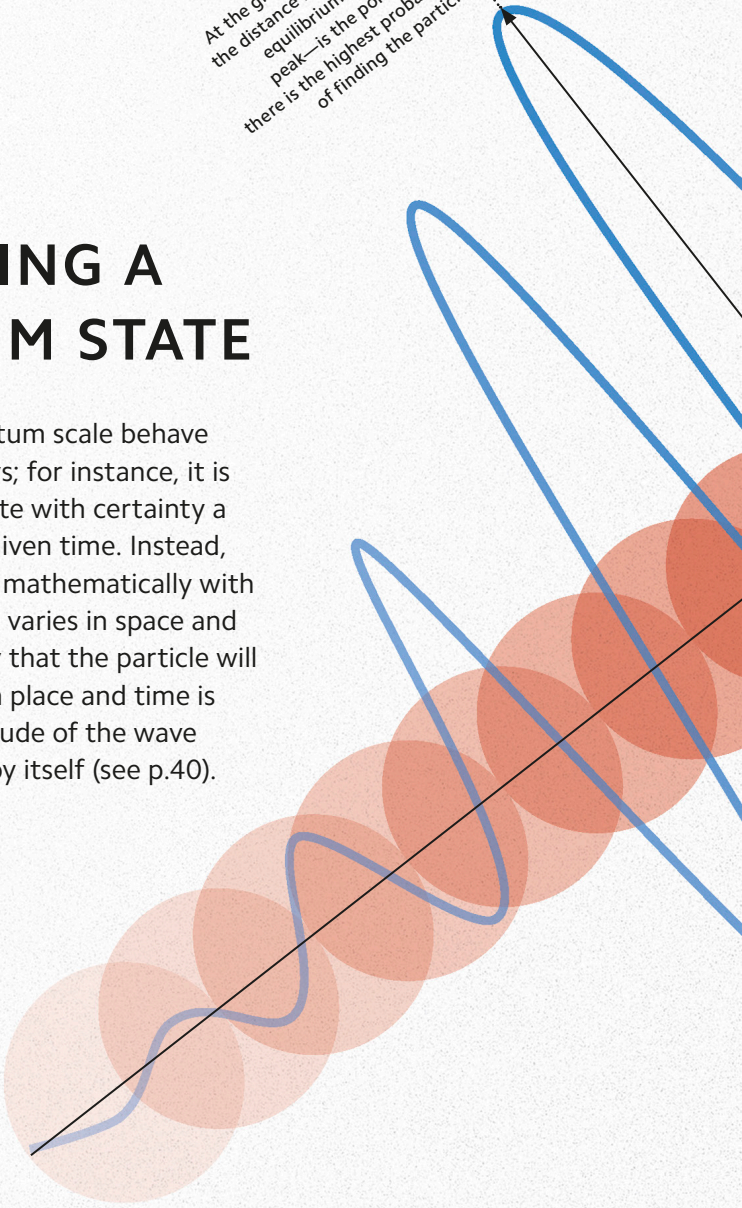


Basic wave function

This image is an example of a wave function for a particle moving in one dimension.

At the greatest amplitude—the distance from the central equilibrium point to the peak—is the highest probability there is of finding the particle.

HIGH PROBABILITY



DESCRIBING A QUANTUM STATE

Objects on the quantum scale behave in unpredictable ways; for instance, it is impossible to calculate with certainty a particle's state at a given time. Instead, its state is described mathematically with a wave function that varies in space and time. The probability that the particle will be found at a certain place and time is related to the amplitude of the wave function multiplied by itself (see p.40).

